

Ismail Aissaoui
Email: i.aissaoui@esi-sba.dz
Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app

Fabrice Gamboa (Univ Toulouse)

Subject: Application for PhD Position – Interpretable and Robust Machine Learning for Physics-Based Numerical Simulations

Dear Fabrice Gamboa,

I am applying for the PhD position on "Interpretable and Robust Machine Learning for Physics-Based Numerical Simulations" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

My deep expertise in Scientific Machine Learning (SciML) and Explainable AI is highly relevant to this project. I have directly integrated physics-based simulations (such as the Navier-Stokes equations) with deep learning architectures to build predictive digital twins. I am highly motivated to advance this integration by focusing on the robustness and mathematical interpretability of these hybrid models, ensuring they remain reliable under edge-case physical conditions.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui